[FORENSIC PSYCHOLINGUISTIC & COGNITIVE RISK ENGINE]

SUBTEXT ENGINE

An algorithmic forensic engine for detecting covert manipulation, systemic deception, reflexive control, and testimony fabrication across high-stakes negotiations, private equity due diligence, and elite corporate litigation.

21

PEER-REVIEWED SCIENTIFIC TREATISES (FBI, BGH 1999, CAMBRIDGE, LIWC, RAS)

8

GLOBAL FORENSIC DISCIPLINES (STATEMENT ANALYSIS, CBCA, REFLEXIVE CONTROL)

0%

DATA LEAKAGE: AIR-GAPPED PRIVATE VPC & ON-PREMISE LOCAL DEPLOYMENT

100%

COURT-ADMISSIBLE METRICS: STRICTLY EXCLUDES DISCREDITED COMMERCIAL SCAN

ARCHITECTURAL DEFECT // FRONTIER LLM EVALUATION

THE DANGEROUS BLIND SPOT OF GENERATIVE AI

Why standard LLMs and traditional due diligence fail to detect sophisticated deception

EVALUATION VECTOR: CONTENT VS. STRUCTURE
DEFECT: RLHF SYCOPHANCY

1. THE CONTENT-LAYER VULNERABILITY

Standard LLMs (GPT-4o, Claude 3.5, Gemini) and conventional legal teams evaluate communication at the **semantic content layer** (*what* is said). Sophisticated bad-faith counterparties exploit this vulnerability by producing syntactically pristine narratives that conceal catastrophic liabilities beneath plausible rhetoric.

2. RLHF-INDUCED SYCOPHANCY

Reinforcement Learning from Human Feedback (RLHF) optimizes commercial AI models to be agreeable, harmless, and polite. When presented with evasive executive testimony or high-stakes contracts, frontier LLMs default to "charitable interpretation," actively rationalizing glaring temporal gaps and psychological red flags.

3. THE SUBTEXT ENGINE SOLUTION

Subtext Engine operates exclusively at the **structural, syntactic, and reflexive layers**: *how* claims are phrased, *what* critical experiential details are deleted, and *how* the decision-maker's mental filter is being manipulated. We replace superficial sentiment analysis with verifiable forensic psycholinguistics.

SCIENTIFIC FOUNDATION // 21 TREATISES

THE 8 GLOBAL FORENSIC INTELLIGENCE PILLARS

A mathematically unified framework validated by German, European, and US judicial precedents

EVIDENTIARY STANDARDS: **BGH 1999 // DAUBERT // FRYE**

01 // STATEMENT ANALYSIS

STATEMENT BALANCE & SYNTAX

FBI 20/50/30 structural balance, 1st-person pronoun drop, reliable denials ("I did not [X]"), and temporal lacunae (Adams, McClish, Coulthard, Olsson).

02 // JUDICIAL CREDIBILITY

CBCA & REALITY MONITORING

19 Undeutsch criteria of experiential memory (German Federal Court of Justice BGH 1999 standard). Cognitive load elevation in deceit (Steller, Köhnken, Vrij).

03 // COMPUTATIONAL METRICS

QUANTITATIVE LIWC & LDI RATIOS

Mathematical function word ratios, exclusive particle dropping, and the Linguistic Distancing Index (LDI) for psychological detachment (Pennebaker, Newman).

04 // CYBERNETICS & DEFENSE

REFLEXIVE CONTROL & COERCION

Lefebvre's reflexive polynomials, false double-binds, Hassan's B.I.T.E. model, and Biderman's UN interrogation manipulation matrices.

Complete Stack: Cognitive Alethiology (Sartori aIAT), Affective Mimicry & Remorse (Porter), Global Evaluation System (Arce SEG), and Cross-Cultural SUE (Nortje).

21 FOUNDATIONAL TREATISES

FOUR HIGH-STAKES COMMERCIAL ARENAS

Where subtle deception, narrative inflation, and hidden agendas cost millions of dollars

APPLICATION: **HIGH-VALUE TRANSACTIONS**

1. PRIVATE EQUITY & M&A DUE DILIGENCE

INVESTMENT & M&A

- **Founder Authenticity Audits:** Screening executive interviews, pitch transcripts, and data room memos for unacknowledged liabilities.
- **Narrative Inflation Detection:** Pinpointing linguistic markers of exaggerated ARR retention and customer churn rationalization.
- **Shadow Negotiation Profiling:** Uncovering counterparty BATNA, hidden friction points, and deal-breaking redlines before signing.

2. CORPORATE INVESTIGATIONS & FORENSIC RISK

FORENSIC RISK & GRC

- **Whistleblower Credibility Scoring:** Filtering genuine multi-million-dollar corruption tips from fabricated competitor smears using CBCA.
- **Procurement Fraud Collusion:** Detecting terminology mutation across ostensibly independent supplier tender bids.
- **Internal Fraud Interviews:** Assisting forensic accountants and corporate investigators in identifying rehearsed alibis and deceptive evasion.

3. ELITE WHITE-COLLAR DEFENSE & ARBITRATION

BIG LAW & ARBITRATION

- **Deposition Preparation:** Stress-testing key witness statements under Coulthard & Olsson forensic linguistic models before depositions.
- **International Arbitration (LCIA, ICC, SIAC):** Auditing written witness statements for institutional coaching and syntax stalls.
- **Expert Witness Submissions:** Providing objective, methodology-backed forensic psycholinguistic reports admissible under Daubert.

4. ENTERPRISE SECURITY & COUNTER-REFLEXIVE INTELLIGENCE

GOVTECH & DEFENSE

- **Insider Threat Early Warning:** Monitoring communicative anomalies, linguistic distancing shifts, and coercion markers in sensitive channels.
- **Countering Reflexive Manipulation:** Shielding corporate boards and negotiating teams from adversarial state/competitor influence campaigns.
- **Hostile Negotiation Shield:** Real-time linguistic telemetry during high-tension extortion crises or regulatory investigations.

MARKET LANDSCAPE // GLOBAL ENTERPRISE MATRIX

TOP 20 GLOBAL TARGET INSTITUTIONS: ENTRY VECTORS

Decision-makers, critical pain points, and specific Subtext Engine integration vectors

ADDRESSABLE MARKET: **$12.4B ENTERPRISE RISK & LITIGATION**

INSTITUTION / GLOBAL TIER	TARGET DECISION-MAKER (BUYER)	CRITICAL VULNERABILITY & COST OF FAILURE	SUBTEXT ENGINE SOLUTION VECTOR
Kroll / FTI Consulting / Control Risks / Alvarez & Marsal FORENSIC INVESTIGATIONS	Global Heads of Investigations, Forensic Risk Managing Directors, Senior Partners	Rehearsed alibis, hidden asset transfers, subjective investigator bias, unverified whistleblower tips costing clients $50M+.	Automated CBCA / Statement Analysis scoring of interview transcripts and confidential informant dossiers.
Kirkland & Ellis / Latham & Watkins / Skadden / Quinn Emanuel BIG LAW WHITE-COLLAR DEFENSE	Chairs of White-Collar Defense, Senior Litigation Partners, International Arbitration Heads	Coached witness testimony unraveling on the stand; unprovable concealed intent in complex civil fraud or SEC enforcement.	Deposition stress-testing; institutional syntax contamination audit; forensic linguistic expert evidence reports.
Blackstone / KKR / Carlyle Group / Insight Partners PRIVATE EQUITY & GROWTH CAPITAL	Managing Directors, Operating Partners, Due Diligence Heads, Investment Committee Members	Investing $200M+ based on fabricated metrics, inflated customer satisfaction narratives, or concealed founder IP disputes.	Linguistic Distancing Index (LDI) and deceptive omission audits on founder management presentations and Q&A transcripts.
Palantir Technologies / Booz Allen Hamilton / Dataminr DEFENSE & THREAT INTELLIGENCE	Chief Risk Officers, Threat Intelligence Directors, Heads of Counter-Intelligence Practices	Adversarial reflexive control operations; cognitive deception campaigns by state-backed actors; malicious insider threats.	Integration of Lefebvre reflexive polynomial algorithms into enterprise threat intelligence pipelines.
McKinsey & Co. / Bain & Co. (PE Practice) / BCG STRATEGIC MANAGEMENT ADVISORY	Senior Partners (Restructuring & M&A), Heads of Strategic Risk, Board Advisory Leads	Advising Fortune 500 boards on cross-border joint ventures while blind to counterparty bad-faith negotiating tactics.	Shadow negotiation profiling; real-time cognitive load mapping during cross-border M&A negotiations.

FORENSIC CASE STUDY // LIVE DECONSTRUCTION

ANATOMY OF DECEPTION: A $120M M&A DEPOSITION

Contrasting human surface review against the Subtext Engine structural X-Ray

DISCIPLINE: **M&A DUE DILIGENCE AUDIT**

RAW FOUNDER TESTIMONY (M&A LITIGATION)

> *"I was actively leading our enterprise expansion in Q3. We diligently executed all tier-one customer onboarding protocols. There were no material cancellations reported to me personally. Around mid-August, the platform upgrade was deployed. Everything was operating according to roadmap specifications. A decision was made to postpone enterprise reporting due to ongoing architectural scaling. I would never misrepresent our core churn metrics to incoming private equity investors."*

Typical Corporate Due Diligence Assessment:
"Founder exudes confidence, accepts leadership role, explicitly denies customer cancellations, and defends company integrity."

SUBTEXT ENGINE FORENSIC VERDICT

🔴 **1. Pronoun Evaporation & Responsibility Severing:** Moves from personal agency (*"I was leading"*) to passive evasion (*"A decision was made to postpone"*). Sudden agent deletion at the precise point of failure marks institutional guilt.

🔴 **2. Equivocal Subjunctive Denial:** *"I would never misrepresent..."* is a hypothetical non-denial, NOT a reliable factual denial (*"I did not misrepresent"*). McClish Statement Law: innocent subjects make direct factual assertions in the simple past tense.

🔴 **3. Subjective Memory Void (CBCA Criteria 3 & 4):** Zero perceptual, spontaneous, or sensory details regarding the mid-August upgrade; high cognitive friction around customer retention numbers.

⚠️ **Strategic Verdict: Subpoena internal Slack communications between Aug 10–25. Founder was fully aware of 40% enterprise contract churn prior to closing.**

OUTCOME: THE INVESTMENT COMMITTEE AVOIDED A $120M WRITE-DOWN AFTER CONFIRMING CONTRACT CHURN

COMMERCIAL ENGAGEMENT // DEPLOYMENT MODELS

ENTERPRISE DEPLOYMENT & PARTNERSHIP MODELS

From bespoke high-stakes shadow advisory to enterprise air-gapped on-premise integration

SLA: AIR-GAPPED // ZERO TELEMETRY LEAKAGE

TIER 01

FORENSIC ADVISORY & SHADOW DESK

$25K – $75K / deal

Bespoke, rapid-response forensic profiling for active M&A transactions, hostile takeovers, or emergency whistleblower crises under strict attorney-client privilege.

- 24–48 hour turnaround on dossiers
- Counterparty shadow-profiling
- Pre-deposition testimony auditing

TIER 02 (FLAGSHIP)

AIR-GAPPED PRIVATE VPC DEPLOYMENT

$120K – $350K / yr

Full enterprise integration within your isolated corporate cloud (AWS GovCloud / Azure Dedicated / On-Premise bare metal). Zero external API calls, zero telemetry leakage.

- Local weights (Llama 3 / Mistral / DeepSeek)
- Batch processing of 100K+ legal discovery docs
- Enterprise SLA, custom API & prompt fine-tuning

TIER 03

OEM & LEGALTECH PLATFORM LICENSE

Custom Royalty

Embed Subtext Engine's 21 algorithms directly into your proprietary GRC platform, e-Discovery software, or ATS HR background verification ecosystem.

- White-label API & SDK libraries
- Full access to raw psycholinguistic matrices
- Co-branded academic validation credentials

RETURN ON INVESTMENT: PREVENTING A SINGLE TOXIC EXECUTIVE HIRE OR FRAUDULENT ACQUISITION DELIVERS INFINITE ROI

PROVENANCE & ENGAGEMENT // NEXT STEPS

INITIATE A CONFIDENTIAL PROOF OF CONCEPT

Evaluate Subtext Engine in a blinded trial using your organization's historical resolved cases

NEXT STEP: **BLINDED TRIAL PROTOCOL**

THE BLINDED PROOF OF CONCEPT (POC) PROTOCOL

Provide us with 2–3 anonymized, historical investigative texts (fraud depositions, contested whistleblower reports, or negotiation transcripts where the true factual outcome is already known to your security committee).

Our engine will generate a blinded forensic teardown report. Compare our structural psycholinguistic conclusions against ground truth to confirm the engine's diagnostic fidelity firsthand.

```
$ agy install elevchuk-lab/subtext-engine
```

LABORATORY CREDENTIALS & EXECUTIVE CONTACT

- 👤 **Yevgeniy Levchuk** (Principal Investigator, elevchuk-lab)
- 🏛️ **Research Focus:** Computational Cognitive Forensics & Deception Detection
- 🔗 **Open Source Core:** github.com/elevchuk-lab/subtext-engine
- 📚 **Academic Registry:** ORCID Verified // CERN Zenodo Open Access
- 📜 **Licensing:** MIT Open Source Core / Enterprise Commercial Dual-License

* The core engine is fully open-source for academic peer-review, security vetting, and algorithmic transparency prior to enterprise deployment.